

DD α AMG for the Schwinger model

We document and explain in detail the implementation of the DD α AMG solver for the 2-flavor Schwinger model with a twisted mass term and the clover improvement.

1 Schwinger model

The Schwinger model represents Quantum Electrodynamics (QED) in two dimensions: one space and one time dimension. The model shares similar properties with QCD, such as confinement, chiral symmetry breaking and their topology. Mathematically, the Schwinger model has less degrees. For these reasons, it is a good *toy model* for testing numerical methods which could be applied to QCD as well. We define the Dirac matrix for the Schwinger model in this section.

Let us define the $2d$ lattice as

$$\Lambda := \{n = (n_t, n_x) \mid n_t = 0, 1, \dots, N_t - 1; n_x = 0, 1, \dots, N_x - 1\}. \quad (1)$$

At each site there are two *spin* components, *i.e.* at each site we consider two complex numbers. Therefore, a vector defined on the lattice has the form $\psi_\mu(n)$, where $\mu = 0, 1$ and $n \in \Lambda$. Then, we have a total of $2N_x N_t$ complex variables. Thus, the set of variables is

$$\mathcal{V} = \Lambda \times \{0, 1\}. \quad (2)$$

The Dirac-Wilson operator (in lattice units) is defined as

$$\begin{aligned} D_{\alpha\beta}[n, n'] &= (m_0 + 2)\delta^{\alpha\beta}\delta_{n,n'} \\ &- \frac{1}{2} \sum_{\mu=0}^1 \left[(1 - \sigma_\mu)^{\alpha\beta} U_\mu(n) \delta_{n+\hat{\mu}, n'} + (1 + \sigma_\mu)^{\alpha\beta} U_\mu^\dagger(n - \hat{\mu}) \delta_{n-\hat{\mu}, n'} \right]. \end{aligned} \quad (3)$$

The index $\mu = 0$ refers to time and $\mu = 1$ to space and m_0 is the *bare mass*. The vector $\hat{\mu}$ represents a unit vector in the direction indexed by μ . We choose the following representation for σ

$$\sigma_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_1 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \quad (4)$$

The *link variables* $U_\mu(n) \in \text{U}(1)$ are randomly generated during simulations. We consider periodic boundaries for the space direction and antiperiodic for the time direction, *i.e.* for any field $\psi_\mu(n)$ on the lattice we have

$$\psi_\mu[(N_x - 1 + 1, n_t)] = \psi_\mu[(0, n_t)], \quad \psi_\mu[(n_x, N_t - 1 + 1)] = -\psi_\mu[(n_x, 0)]. \quad (5)$$

The application of D to a field on the lattice is

$$(D\psi)_\alpha(n) = (m_0 + 2)\psi_\alpha(n) - \frac{1}{2} \sum_{\mu, \beta=0}^1 \left[(1 - \sigma_\mu)^{\alpha\beta} U_\mu(n) \psi_\beta(n + \hat{\mu}) + (1 + \sigma_\mu)^{\alpha\beta} U_\mu^\dagger(n - \hat{\mu}) \psi_\beta(n - \hat{\mu}) \right]. \quad (6)$$

Highly ill-conditioned systems arise when we simulate close to criticality. In this model, this means when the physical fermion mass vanishes. Such a behavior is obtained when m_0 and β (bare mass and gauge coupling) take the values in Table 1, see Ref. [1]. We run simulations with parameters close to the ones in Table 1 for testing our method.

β	$-m_{\text{crit}}$
2	0.1968(9)
3	0.1351(2)
4	0.1033(1)
5	0.0840(1)
6	0.0719(1)

Table 1: Critical values for the Schwinger model [1].

We implement the multilevel preconditioner to invert the Dirac matrix introduced in Ref. [2], but adapted to the Schwinger model. The preconditioner is fed into the FGMRES method to solve the linear problem $Dx = \psi$. For smoothing the system we use *Schwarz Alternating Procedure* (SAP), a domain decomposition method. On the coarsest level the solver is GMRES (no preconditioning) with a tolerance of 10^{-2} . The construction of the interpolator P is done with an aggregation based approach, which we explain here. We use either a K- or V-cycling strategy.

2 Schwarz Alternating Procedure (SAP)

SAP is a domain decomposition method. Its general idea is based on dividing the lattice into blocks $\mathcal{L}_i$, each one containing a different set of variables $\mathcal{V}_i$. See for instance Figure 1. By restricting the application of the Dirac operator to these blocks, we can invert the local systems approximately and use them to update the global solution. We explain the implementation without discussing theoretical details, which are found in Refs. [2, 3].

We restrict the application of the Dirac matrix to the blocks with an operator

$$I_{\mathcal{V}} : \mathcal{V}_i \rightarrow \mathcal{V}, \quad (7)$$

where $\mathcal{V}$ corresponds to the whole set of variables and $\mathcal{V}_i \subset \mathcal{V}$ to a subset. We define the local operators

$$D_i := I_{\mathcal{V}_i}^T D I_{\mathcal{V}_i} \quad \text{and} \quad A_i := I_{\mathcal{V}_i} D_i^{-1} I_{\mathcal{V}_i}^T. \quad (8)$$

We use the solutions of the local systems

$$D_i e_i = I_{\mathcal{V}_i}^T r, \quad r = \psi - Dx \quad (9)$$

to update the global solution x . Since D only considers nearest-neighbour coupling, the blocks can be split into two sets forming a checkerboard pattern (red/black decomposition). In this manner, we

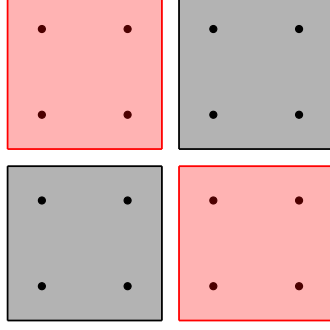


Figure 1: Red/Black decomposition of the lattice. Each point inside the blocks has two complex degrees of freedom.

can solve equations 9 simultaneously for all the red or black blocks. This allows for parallelization. We define one SAP iteration as

$$x \leftarrow (I - KD)x + K\psi, \quad (10)$$

where

$$K = A_{\text{black}}(I - DA_{\text{red}}) + A_{\text{red}}, \quad A_{\text{color}} = \sum_{i \in \text{color}} A_i. \quad (11)$$

In practice, SAP is implemented as we show in Algorithm 1. The restricted Dirac operator is the same as eq. (3), but only considering the sites that belong to the same block

$$\begin{aligned} (D_{\alpha\beta}[n, n'])_{\mathcal{L}_i} &= (m_0 + 2)\delta^{\alpha\beta}\delta_{n, n'} \\ &\quad - \frac{1}{2} \sum_{\mu=0}^1 \left[(1 - \sigma_\mu)^{\alpha\beta} U_\mu(n) \delta_{n+\hat{\mu}, n'} + (1 + \sigma_\mu)^{\alpha\beta} U_\mu^\dagger(n - \hat{\mu}) \delta_{n-\hat{\mu}, n'} \right], \\ n, n', n + \hat{\mu}, n - \hat{\mu} &\in \mathcal{L}_i. \end{aligned} \quad (12)$$

If one site is not in $\mathcal{L}_i$, we fix the corresponding term to zero. To invert the local systems we use GMRES to achieve a relative tolerance of 10^{-3} . This low-precision solution makes SAP a non-stationary method which is not guaranteed to converge as a stand-alone solver when the matrix is very ill-conditioned. For that reason, SAP is better used as a preconditioner or smoother. For DD α AMG, SAP is used as the latter for every grid level.

Algorithm 1 Schwarz Alternating Procedure

Inputs: ψ, b . Output: ψ

$r = \psi - D\psi$

for $i = 1, \dots, \nu$:

 for all red i :

$\psi \leftarrow \psi + A_i^{-1}r$

$r = \psi - D\psi$

 for all black i :

$\psi \leftarrow \psi + A_i^{-1}r$

$r = \psi - D\psi$

3 Aggregation based approach

DD α is mainly characterized by the construction of its interpolator P . This follows an aggregation approach, which is based on the lattice geometry, and a set of *test vectors* $\{w_i | i = 1, \dots, N_v\}$. These vectors should capture the *null-space* to improve the convergence properties of the method. They are obtained during a setup phase, which we describe in this section. The coarse grid matrix is computed with the Galerkin product $D_c = P^\dagger D P$.

To build the aggregates, we first decompose the lattice into N_B blocks. The exact definition for the blocking is in Definition 2.2 of Ref. [2], but for the sake of simplicity we explain the idea with an example. Let us consider a 6^2 volume. We call `block_x` and `block_t` to the number of sites contained in each block in the x and t directions respectively. We have the freedom to choose the value of `block_i` as long as N_i/block_i is an integer number. For instance, if `block_x`=2 and `block_t`=3 we have six lattice blocks

$$\begin{aligned}\mathcal{L}_1 &= \{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2)\}, \\ \mathcal{L}_2 &= \{(0, 3), (0, 4), (0, 5), (1, 3), (1, 4), (1, 5)\}, \\ \mathcal{L}_3 &= \{(2, 0), (2, 1), (2, 2), (3, 0), (3, 1), (3, 2)\}, \\ \mathcal{L}_4 &= \{(2, 3), (2, 4), (2, 5), (3, 3), (3, 4), (3, 5)\}, \\ \mathcal{L}_5 &= \{(4, 0), (4, 1), (4, 2), (5, 0), (5, 1), (5, 2)\}, \\ \mathcal{L}_6 &= \{(4, 3), (4, 4), (4, 5), (5, 3), (5, 4), (5, 5)\}.\end{aligned}$$

The ordered pairs are of the form (n_t, n_x) . Other blockings can be built in a similar manner by taking cartesian products. In order to build the aggregates we also consider the spin component μ . For each lattice block we define two aggregates: one for $\mu = 0$ and another one for $\mu = 1$, that is

$$\mathcal{A}_{2i-1} = \mathcal{L}_i \times \{0\}, \quad \mathcal{A}_{2i} = \mathcal{L}_i \times \{1\}. \quad (13)$$

Thus, the number of aggregates is twice the number of lattice blocks, *i.e.* $N_{\mathcal{A}} = 2N_B$. The blocking used for the aggregates does not need to coincide with the blocking for SAP.

We now need a set of test vectors to build the interpolator. We describe in a merely practical way how to obtain them:

- Generate a set of N_v random vectors of dimension $2N_x N_t$. We denote the vectors as w_i .
- Approximately solve $Dx = 0$ with initial solution w_i for every test vector. Update $w_i \leftarrow x$. A couple of iterations with the smoother is enough. The idea is to make the test vectors rich in near-kernel components.
- Split the test vectors over the aggregates. We explain the idea. Let us say we have $N_{\mathcal{A}}$ aggregates. We restrict w_i by leaving those components that belong to an aggregate $\mathcal{A}$ unchanged, while zeroing the others. This way we create another $N_v N_{\mathcal{A}}$ vectors of the form

$$\left(w_i^{(j)}\right)_k = \begin{cases} (w_i)_k & \text{if } k \in \mathcal{A}_j, \\ 0 & \text{else,} \end{cases} \quad (14)$$

where $j = 0, \dots, N_{\mathcal{A}} - 1$, $i = 0, \dots, N_v - 1$ and $k = 0, \dots, 2N_x N_t - 1$. Afterwards, the new vectors are orthonormalized. We do the orthonormalization only among those vectors that

share the same index j , *i.e.* only among those vectors restricted to the same aggregate. The interpolator is built by arranging the chopped vectors in columns

$$P = \left(w_1^{(1)} \middle| w_2^{(1)} \middle| \cdots \middle| w_{N_v}^{(1)} \middle| \cdots \middle| w_1^{(N_A)} \middle| w_2^{(N_A)} \middle| \cdots \middle| w_{N_v}^{(N_A)} \right). \quad (15)$$

If the operator is assembled correctly, the sets

$$V_j = \{w_1^{(j)}, w_2^{(j)}, \dots, w_{N_v}^{(j)}\}, \quad j = 1, \dots, N_A \quad (16)$$

should be orthonormal.

- The quality of the interpolator is further improved with *bootstrap adaptivity*. That is, we use the current multilevel method defined by P to approximately solve $Dx = (w_i - Dw_i)$ and update $w_i \leftarrow x$. The resultant test vectors are orthonormalized again as in the previous step.

With these ingredients we can implement the multilevel method. Still, it is convenient to find an explicit representation for the coarse grid operator D_c to avoid calling P and $P^\dagger$ several times during the computation. We show how to do that in the next section. We introduce a new set of coarse variables which allow us to rewrite D_c in an analogue way as the original Dirac matrix.

For more than two levels we repeat the same steps: aggregate the coarse lattice and generate a set of test vectors which are later orthonormalized.

4 Constructing the coarse grid operator

The Dirac operator on the fine grid, eq. (3), is defined by the gauge links and considering nearest-neighbor couplings. On coarser levels, we can preserve a similar structure by introducing a set of *coarse gauge links* in such a way that

$$(D_c)_{xy} = [(m_0 + 2) - A(x)] \delta_{xy} - \sum_{\mu=0}^1 (B_\mu(x) \delta_{x+\hat{\mu},y} + C_\mu(x) \delta_{x-\hat{\mu},y}), \quad (17)$$

where x, y are lattice blocks and A, B_μ, C_μ represent the coarse links. On the coarse grid, each point on the lattice acquires $2N_v$ degrees of freedom. We show how to find an explicit representation for $A(x), B_\mu(x)$ and $C_\mu(x)$ for the two-level case. Later we show the generalization for the multilevel method.

We first write the components of P

$$P_{ij} = [w_c^{(x,\alpha)}]_i, \quad (18)$$

where c, α and x represent the test vector, spin and lattice block respectively

$$c = 0, \dots, N_v - 1, \quad \alpha = 0, 1, \quad x = 0, \dots, N_B - 1. \quad (19)$$

We can relate these numbers with the vectorized index j through

$$c = j \bmod N_v, \quad a = \text{int}(j/N_v), \quad x = \text{int}(a/2), \quad \alpha = a \bmod 2, \quad (20)$$

where a is the aggregate associated to j . On the other hand, the vectorized index i represents a variable on the fine grid (n, α) (lattice site and spin). This allows us to write

$$P_{ij} = \left[w_c^{(x,\alpha)} \right]_\alpha (n), \quad i = 2n + \alpha, \quad j = N_v a + c, \quad (21)$$

where $i = 0, \dots, 2N_x N_t - 1$ and $j = 0, \dots, N_{\mathcal{A}} N_v - 1$. In a similar manner, we have

$$P_{ji}^\dagger = \left[w_c^{*(x,\alpha)} \right]_\alpha (n), \quad i = 2n + \alpha, \quad j = N_v a + c. \quad (22)$$

Since we split the test vectors over the aggregates, the following is satisfied

$$\text{if } (n, \alpha) \notin \mathcal{A}_{(x,\alpha)} \implies \left[w_c^{(x,\alpha)} \right]_\alpha (n) = 0, \quad (23)$$

where (x, α) defines the aggregate $\mathcal{A}_{(x,\alpha)} = x \times \{\alpha\}$.

Let us find an expression for $(P^\dagger DP)_{ij}$ with this information

$$\begin{aligned} (D_c)_{ij} &= (P^\dagger DP)_{ij} \quad i, j = 0, \dots, N_{\mathcal{A}} N_v - 1 \\ &= \sum_{kl} P_{ik}^\dagger D_{kl} P_{lj} = \sum_{kl} P_{ik}^\dagger D_{kl} \left[w_b^{(y,\beta)} \right]_l = \sum_k P_{ik}^\dagger \left(D w_b^{(y,\beta)} \right)_k \end{aligned} \quad (24)$$

The indices y, β and b are related to j according to eq. (20) and l represents a variable on the fine grid. We use eq. (6) to simplify the application of D over $w_b^{(y,\beta)}$

$$\begin{aligned} \left[D w_b^{(y,\beta)} \right]_\gamma (n) &= (m_0 + 2) \left[w_b^{(y,\beta)} \right]_\gamma (n) \\ &\quad - \frac{1}{2} \sum_{\mu, \delta=0}^1 \left[(1 - \sigma_\mu)^{\gamma\delta} U_\mu(n) \left[w_b^{(y,\beta)} \right]_\delta (n + \hat{\mu}) + (1 + \sigma_\mu)^{\gamma\delta} U_\mu^\dagger(n - \hat{\mu}) \left[w_b^{(y,\beta)} \right]_\delta (n - \hat{\mu}) \right]. \end{aligned}$$

If we consider that $k = 2\gamma + n$, we can write eq. (24) as

$$\begin{aligned} [D_c]_{cb}^{\alpha\beta}(x, y) &= \sum_{n \in \Lambda, \gamma} (m_0 + 2) \left[w_c^{*(x,\alpha)} \right]_\gamma (n) \left[w_b^{(y,\beta)} \right]_\gamma (n) \\ &\quad - \frac{1}{2} \sum_{n \in \Lambda, \mu, \gamma, \delta=0}^1 \left[w_c^{*(x,\alpha)} \right]_\gamma (n) \left[(1 - \sigma_\mu)^{\gamma\delta} U_\mu(n) \left[w_b^{(y,\beta)} \right]_\delta (n + \hat{\mu}) + (1 + \sigma_\mu)^{\gamma\delta} U_\mu^\dagger(n - \hat{\mu}) \left[w_b^{(y,\beta)} \right]_\delta (n - \hat{\mu}) \right]. \end{aligned}$$

We relate this equation with eq. (24) through

$$\begin{aligned} c &= i \bmod N_v, \quad a = \text{int}(i/N_v), \quad x = \text{int}(a/2), \quad \alpha = a \bmod 2, \\ b &= j \bmod N_v, \quad a' = \text{int}(j/N_v), \quad y = \text{int}(a'/2), \quad \beta = a' \bmod 2, \end{aligned} \quad (25)$$

Due to the local orthonormalization of the test vectors, the first term is different from zero when $x = y, \alpha = \beta$ and $c = b$. For the second term, the test vectors are zero unless $\alpha = \gamma$ and $\beta = \delta$. Thus,

$$\begin{aligned} [D_c]_{cb}^{\alpha\beta}(x, y) &= (m_0 + 2) \delta_{xy} \delta^{\alpha\beta} \delta_{cb} \\ &\quad - \frac{1}{2} \sum_{n \in \Lambda, \mu=0}^1 \left[w_c^{*(x,\alpha)} \right]_\alpha (n) \left[(1 - \sigma_\mu)^{\alpha\beta} U_\mu(n) \left[w_b^{(y,\beta)} \right]_\beta (n + \hat{\mu}) + (1 + \sigma_\mu)^{\alpha\beta} U_\mu^\dagger(n - \hat{\mu}) \left[w_b^{(y,\beta)} \right]_\beta (n - \hat{\mu}) \right]. \end{aligned}$$

The lattice blocking implies that the second term is only different from zero when $y = x, y = x + \hat{\mu}$

or $y = x - \hat{\mu}$. Then, we have

$$\begin{aligned}
[D_c]_{cb}^{\alpha\beta}(x, y) &= (m_0 + 2)\delta_{xy}\delta^{\alpha\beta}\delta_{cb} - \frac{1}{2} \sum_{\mu, n \in x} (1 - \sigma_\mu)^{\alpha\beta} \left[w_c^{*(x, \alpha)} \right]_\alpha(n) U_\mu(n) \left[w_b^{(x, \beta)} \right]_\beta(n + \hat{\mu}) \\
&\quad - \frac{1}{2} \sum_{\mu, n \in x} (1 - \sigma_\mu)^{\alpha\beta} \left[w_c^{*(x, \alpha)} \right]_\alpha(n) U_\mu(n) \left[w_b^{(x + \hat{\mu}, \beta)} \right]_\beta(n + \hat{\mu}) \\
&\quad - \frac{1}{2} \sum_{\mu, n \in x} (1 + \sigma_\mu)^{\alpha\beta} \left[w_c^{*(x, \alpha)} \right]_\alpha(n) U_\mu^\dagger(n - \hat{\mu}) \left[w_b^{(x, \beta)} \right]_\beta(n - \hat{\mu}) \\
&\quad - \frac{1}{2} \sum_{\mu, n \in x} (1 + \sigma_\mu)^{\alpha\beta} \left[w_c^{*(x, \alpha)} \right]_\alpha(n) U_\mu^\dagger(n - \hat{\mu}) \left[w_b^{(x - \hat{\mu}, \beta)} \right]_\beta(n - \hat{\mu}). \tag{26}
\end{aligned}$$

Let us define

$$\begin{aligned}
[A(x)]_{cb}^{\alpha\beta} &:= \frac{1}{2} \sum_{\mu, n \in x} \left[(1 - \sigma_\mu)^{\alpha\beta} \left[w_c^{*(x, \alpha)} \right]_\alpha(n) U_\mu(n) \left[w_b^{(x, \beta)} \right]_\beta(n + \hat{\mu}) \right. \\
&\quad \left. + (1 + \sigma_\mu)^{\alpha\beta} \left[w_c^{*(x, \alpha)} \right]_\alpha(n) U_\mu^\dagger(n - \hat{\mu}) \left[w_b^{(x, \beta)} \right]_\beta(n + \hat{\mu}) \right] \tag{27}
\end{aligned}$$

$$[B_\mu(x)]_{cb}^{\alpha\beta} := \frac{1}{2} \sum_{n \in x} (1 - \sigma_\mu)^{\alpha\beta} \left[w_c^{*(x, \alpha)} \right]_\alpha(n) U_\mu(n) \left[w_b^{(x + \hat{\mu}, \beta)} \right]_\beta(n + \hat{\mu}) \tag{28}$$

$$[C_\mu(x)]_{cb}^{\alpha\beta} := \frac{1}{2} \sum_{n \in x} (1 + \sigma_\mu)^{\alpha\beta} \left[w_c^{*(x, \alpha)} \right]_\alpha(n) U_\mu^\dagger(n - \hat{\mu}) \left[w_b^{(x - \hat{\mu}, \beta)} \right]_\beta(n - \hat{\mu}). \tag{29}$$

Thus, if we have a vector v on the coarse grid, we can write the application $D_c v$ as

$$\begin{aligned}
(D_c v)_c^\alpha(x) &= (m_0 + 2)v_c^{(x, \alpha)} - \sum_{y, \beta, b} [A(x)]_{cb}^{\alpha\beta} \delta_{xy} v_b^{(y, \beta)} \\
&\quad - \sum_{\mu} \left([B_\mu(x)]_{cb}^{\alpha\beta} \delta_{x + \hat{\mu}, y} v_b^{(y, \beta)} + [C_\mu(x)]_{cb}^{\alpha\beta} \delta_{x - \hat{\mu}, y} v_b^{(y, \beta)} \right). \tag{30}
\end{aligned}$$

Therefore

$$(D_c)_{xy} = [(m_0 + 2) - A(x)] \delta_{xy} - \sum_{\mu=0}^1 [B_\mu(x) \delta_{x + \hat{\mu}, y} + C_\mu(x) \delta_{x - \hat{\mu}, y}], \tag{31}$$

where we assume that a sum over α, β, c, b is performed when multiplied by a vector on the coarse grid. We only build A, B_μ and C_μ once and store them in memory. For the expression above to work, we need to first locally orthonormalize the test vectors.

Now, let us say we want to proceed to the next coarse level. We again have to split the lattice into $\hat{N}_B$ blocks (the previous blocks are now our lattice sites), build $\hat{N}_A = 2\hat{N}_B$ aggregates and a set of $\hat{N}_v$ test vectors which are locally orthonormalized. The components of the interpolator are

$$\hat{P}_{ij} = \left[w_{\hat{c}}^{(\hat{x}, \hat{\alpha})} \right]_{c\alpha}(x). \tag{32}$$

where

$$\begin{aligned}
i &= 0, \dots, N_A N_v - 1, & j &= 0, \dots, \hat{N}_A \hat{N}_v - 1, \\
c &= 0, \dots, N_v - 1, & \alpha &= 0, 1, & x &= 0, \dots, N_B - 1, \\
\hat{c} &= 0, \dots, \hat{N}_v - 1, & \hat{\alpha} &= 0, 1, & \hat{x} &= 0, \dots, \hat{N}_B - 1.
\end{aligned} \tag{33}$$

To build $\hat{D}_c = \hat{P}^\dagger D_c P$ we do a similar procedure as before

$$\begin{aligned}
(\hat{D}_c)_{ij} = [\hat{D}_c]_{\hat{c}\hat{b}}^{\hat{\alpha}\hat{\beta}}(\hat{x}, \hat{y}) &= \sum_{x,c,b,\alpha,\beta} (m_0 + 2) [w_{\hat{c}}^{*(\hat{x},\hat{\alpha})}]_{c\alpha}(x) [w_{\hat{b}}^{(\hat{y},\hat{\beta})}]_{c\alpha}(x) \\
&- \sum_{x,c,b,\alpha,\beta} [w_{\hat{c}}^{*(\hat{x},\hat{\alpha})}]_{c\alpha}(x) [A(x)]_{cb}^{\alpha\beta} [w_{\hat{b}}^{(\hat{y},\hat{\beta})}]_{c\alpha}(x) \\
&- \sum_{\mu,x,c,b,\alpha,\beta} [w_{\hat{c}}^{*(\hat{x},\hat{\alpha})}]_{c\alpha}(x) [B_\mu(x)]_{cb}^{\alpha\beta} [w_{\hat{b}}^{(\hat{y},\hat{\beta})}]_{c\alpha}(x + \hat{\mu}) \\
&- \sum_{\mu,x,c,b,\alpha,\beta} [w_{\hat{c}}^{*(\hat{x},\hat{\alpha})}]_{c\alpha}(x) [C_\mu(x)]_{cb}^{\alpha\beta} [w_{\hat{b}}^{(\hat{y},\hat{\beta})}]_{c\alpha}(x - \hat{\mu}). \tag{34}
\end{aligned}$$

The indices i, j are a vectorization of $(\hat{c}, \hat{\alpha}, \hat{x})$ and $(\hat{b}, \hat{\beta}, \hat{y})$ respectively. Remember that

$$[w_{\hat{b}}^{(\hat{x},\hat{\alpha})}]_{c\alpha}(x) = 0 \quad \text{if} \quad (x, \alpha, c) \notin \mathcal{A}_{(\hat{y},\hat{\beta})}. \tag{35}$$

This, together with the local orthonormalization of the test vectors, imply that the first term is again just $m_0 + 2$ multiplied by deltas. We should also note that unless $\hat{\alpha} = \alpha$ and $\hat{\beta} = \beta$, the test vector components are zero, so we can skip the sums over α, β . In addition, the nearest neighbor coupling and the lattice blocking altogether imply

$$\begin{aligned}
[\hat{D}_c]_{\hat{c}\hat{b}}^{\alpha\beta}(\hat{x}, \hat{y}) &= (m_0 + 2) \delta_{\hat{x}\hat{y}} \delta^{\alpha\beta} \delta_{\hat{c}\hat{b}} - \sum_{x \in \hat{x}, c, b} [w_{\hat{c}}^{*(\hat{x},\alpha)}]_{c\alpha}(x) [A(x)]_{cb}^{\alpha\beta} [w_{\hat{b}}^{(\hat{x},\beta)}]_{b\beta}(x) \delta_{\hat{x}\hat{y}} \\
&- \sum_{\mu, x \in \hat{x}, c, b} [w_{\hat{c}}^{*(\hat{x},\alpha)}]_{c\alpha}(x) [B_\mu(x)]_{cb}^{\alpha\beta} [w_{\hat{b}}^{(\hat{x},\beta)}]_{b\beta}(x + \hat{\mu}) \delta_{\hat{x}\hat{y}} \\
&- \sum_{\mu, x \in \hat{x}, c, b} [w_{\hat{c}}^{*(\hat{x},\alpha)}]_{c\alpha}(x) [C_\mu(x)]_{cb}^{\alpha\beta} [w_{\hat{b}}^{(\hat{x},\beta)}]_{b\beta}(x - \hat{\mu}) \delta_{\hat{x}\hat{y}} \\
&- \sum_{\mu, x \in \hat{x}, c, b} [w_{\hat{c}}^{*(\hat{x},\alpha)}]_{c\alpha}(x) [B_\mu(x)]_{cb}^{\alpha\beta} [w_{\hat{b}}^{(\hat{x}+\hat{\mu},\beta)}]_{b\beta}(x + \hat{\mu}) \delta_{\hat{x}+\hat{\mu},\hat{y}} \\
&- \sum_{\mu, x \in \hat{x}, c, b} [w_{\hat{c}}^{*(\hat{x},\alpha)}]_{c\alpha}(x) [C_\mu(x)]_{cb}^{\alpha\beta} [w_{\hat{b}}^{(\hat{x}-\hat{\mu},\beta)}]_{b\beta}(x - \hat{\mu}) \delta_{\hat{x}-\hat{\mu},\hat{y}}. \tag{36}
\end{aligned}$$

We define

$$\begin{aligned}
[\hat{A}(\hat{x})]_{\hat{c}\hat{b}}^{\alpha\beta} &:= \sum_{x \in \hat{x}, c, b} [w_{\hat{c}}^{*(\hat{x},\alpha)}]_{c\alpha}(x) [A(x)]_{cb}^{\alpha\beta} [w_{\hat{b}}^{(\hat{x},\beta)}]_{b\beta}(x) \\
&+ \sum_{\mu, x \in \hat{x}, c, b} [w_{\hat{c}}^{*(\hat{x},\alpha)}]_{c\alpha}(x) [B_\mu(x)]_{cb}^{\alpha\beta} [w_{\hat{b}}^{(\hat{x},\beta)}]_{b\beta}(x + \hat{\mu}) \\
&+ \sum_{\mu, x \in \hat{x}, c, b} [w_{\hat{c}}^{*(\hat{x},\alpha)}]_{c\alpha}(x) [C_\mu(x)]_{cb}^{\alpha\beta} [w_{\hat{b}}^{(\hat{x},\beta)}]_{b\beta}(x - \hat{\mu}), \\
[\hat{B}_\mu(\hat{x})]_{\hat{c}\hat{b}}^{\alpha\beta} &:= \sum_{\mu, x \in \hat{x}, c, b} [w_{\hat{c}}^{*(\hat{x},\alpha)}]_{c\alpha}(x) [B_\mu(x)]_{cb}^{\alpha\beta} [w_{\hat{b}}^{(\hat{x}+\hat{\mu},\beta)}]_{b\beta}(x + \hat{\mu}), \\
[\hat{C}_\mu(\hat{x})]_{\hat{c}\hat{b}}^{\alpha\beta} &:= \sum_{\mu, x \in \hat{x}, c, b} [w_{\hat{c}}^{*(\hat{x},\alpha)}]_{c\alpha}(x) [C_\mu(x)]_{cb}^{\alpha\beta} [w_{\hat{b}}^{(\hat{x}-\hat{\mu},\beta)}]_{b\beta}(x - \hat{\mu}). \tag{37}
\end{aligned}$$

Thus, the application of $\hat{D}_c$ on a coarse vector $\hat{v}$ is

$$\begin{aligned}
(\hat{D}_c v)_{\hat{c}\alpha}(\hat{x}) &= (m_0 + 2)v_{\hat{c}}^{(\hat{x},\alpha)} - \sum_{\hat{y},\beta,\hat{b}} \left[\hat{A}(\hat{x}) \right]_{\hat{c}\hat{b}}^{\alpha\beta} \delta_{\hat{x}\hat{y}} v_b^{(\hat{y},\beta)} \\
&\quad - \sum_{\mu} \left(\left[\hat{B}_{\mu}(\hat{x}) \right]_{\hat{c}\hat{b}}^{\alpha\beta} \delta_{\hat{x}+\hat{\mu},\hat{y}} v_b^{(\hat{y},\beta)} + \left[\hat{C}_{\mu}(\hat{x}) \right]_{\hat{c}\hat{b}}^{\alpha\beta} \delta_{\hat{x}-\hat{\mu},\hat{y}} v_b^{(\hat{y},\beta)} \right). \tag{38}
\end{aligned}$$

Therefore, the operator $\hat{D}_c$ also has the form

$$(\hat{D}_c)_{\hat{x}\hat{y}} = \left[(m_0 + 2) - \hat{A}(\hat{x}) \right] \delta_{\hat{x}\hat{y}} - \sum_{\mu=0}^1 \left[\hat{B}_{\mu}(\hat{x}) \delta_{\hat{x}+\hat{\mu},\hat{y}} + \hat{C}_{\mu}(\hat{x}) \delta_{\hat{x}-\hat{\mu},\hat{y}} \right]. \tag{39}$$

Clearly, we can repeat the same process to go to the next level and we will again find the same form for the operator, with the same definition for the coarse links in eq. (37). In fact, we can even use the same expression for the finest level if we define

$$[A(x)]_{cb}^{\alpha\beta} := 0, [B_{\mu}(x)]_{cb}^{\alpha\beta} := \frac{1}{2}(1 - \sigma_{\mu})^{\alpha\beta} U_{\mu}(x), [C_{\mu}(x)]_{cb}^{\alpha\beta} := \frac{1}{2}(1 + \sigma_{\mu})^{\alpha\beta} U_{\mu}^{\dagger}(x - \hat{\mu}). \tag{40}$$

In that case c and b are spurious. This way, we can always use eq. (37) to form the coarse gauge links to proceed to the next multigrid level. As a final note, the minus sign of the anti-periodic boundary conditions is taken into account in $\delta_{x+\hat{\mu},\hat{y}}$ and $\delta_{x-\hat{\mu},\hat{y}}$, not in the coarse links A, B_{μ}, C_{μ} .

5 Introducing the twisted mass term

The twisted mass term opens the spectrum of the Dirac operator by adding

$$D_{tm} = i\mu_0 \gamma_5 \tau^3, \tag{41}$$

where τ_3 is the third Pauli matrix acting on flavor space. As a consequence, the Dirac operator is split for each flavor

$$D_u = D_W + i\gamma_5 \mu_0, \tag{42}$$

$$D_d = D_W - i\gamma_5 \mu_0. \tag{43}$$

Since we can relate

$$D_d = \gamma_5 D_u^{\dagger} \gamma_5, \tag{44}$$

it is enough to implement only one version. We will do it for u . The twisted mass term is diagonal and its sign depends on γ_5 . Essentially, we only have to modify

$$(m_0 + 2)\delta_{\alpha\beta}\delta_{n,n'} \longrightarrow [(m_0 + 2)\delta_{\alpha\beta} + i\mu_0(\gamma_5)_{\alpha\beta}] \delta_{n,n'}, \tag{45}$$

where

$$(\gamma_5)_{\alpha\beta} = (-1)^{\alpha}\delta_{\alpha\beta}, \quad (\text{No Einstein summation}). \tag{46}$$

On the coarse grid operator the correction is the same as on the fine level, since

$$[(D_c)_{tm}]_{cb}^{\alpha\beta}(x, y) = (P^{\dagger} D_{tm} P)_{cb}^{\alpha\beta}(x, y) = i\mu_0 \sum_{n,\gamma} \left[\omega_c^{*(x,\alpha)} \right]_{\gamma}(n) (\gamma_5)_{\gamma\delta} \left[\omega_b^{(y,\beta)} \right]_{\delta}(n). \tag{47}$$

The term is zero unless $\alpha = \gamma, \beta = \delta$,

$$[(D_c)_{tm}]_{cb}^{\alpha\beta}(x, y) = (P^\dagger D_{tm} P)_{cb}^{\alpha\beta}(x, y) = i\mu_0 \sum_{n, \gamma} \left[\omega_c^{*(x, \alpha)} \right]_\alpha (n) (\gamma_5)_{\alpha\beta} \left[\omega_b^{(y, \beta)} \right]_\beta (n) = (-1)^\alpha \delta_{\alpha\beta} = \gamma_5 \alpha\beta, \quad (48)$$

where we used the local orthonormalization of the test vectors and substituted the specific values of γ_5 . The expression for $(D_c)_{tm}$ is a consequence of the γ_5 symmetry of the interpolator,

$$P\gamma_5 = \gamma_5 P \quad \rightarrow \quad P^\dagger i\mu_0 \gamma_5 P = P^\dagger P i\mu_0 \gamma_5 = i\mu_0 \gamma_5. \quad (49)$$

This, in order to update the whole coarse grid operator it is enough to change

$$[(m_0 + 2) - A(x)] \delta_{xy} \quad \rightarrow \quad [(m_0 + 2) + i\mu_0 \gamma_5 - A(x)] \delta_{xy}. \quad (50)$$

6 Clover term

The clover term is defined by

$$D_{\text{SW}} = \frac{c_{\text{sw}}}{2} \sum_{\mu < \nu} \sigma_{\mu\nu} F_{\mu\nu}, \quad (51)$$

where

$$\sigma_{\mu\nu} = \frac{1}{2i} [\sigma_\mu, \sigma_\nu], \quad F_{\mu\nu} = -\frac{i}{8} (Q_{\mu\nu} - Q_{\mu\nu}^\dagger), \quad (52)$$

and

$$\begin{aligned} Q_{\mu\nu}(x) = & U_\mu(x) U_\nu(x + \hat{\mu}) U_\mu^\dagger(x + \hat{\nu}) U_\nu^\dagger(x) \\ & + U_\nu(x) U_\mu^\dagger(x - \hat{\mu} + \hat{\nu}) U_\nu^\dagger(x - \hat{\mu}) U_\mu(x - \hat{\mu}) \\ & + U_\mu^\dagger(x - \hat{\mu}) U_\nu^\dagger(x - \hat{\mu} - \hat{\nu}) U_\mu(x - \hat{\mu} - \hat{\nu}) U_\nu(x - \hat{\nu}) \\ & + U_\nu^\dagger(x - \hat{\nu}) U_\mu(x - \hat{\nu}) U_\nu(x + \hat{\mu} - \hat{\nu}) U_\mu^\dagger(x). \end{aligned} \quad (53)$$

The function $Q_{\mu\nu}(x)$ is the sum over all plaquettes in the $\mu\nu$ plane touching the site x , taken with counterclockwise orientation. In the Schwinger model we can further reduce this term to obtain

$$D_{\text{SW}}^{\alpha\beta}[x, y] = i \frac{c_{\text{sw}}}{16} (\gamma_5)_{\alpha\beta} (Q_{01}(x) - Q_{01}^\dagger(x)) \delta_{xy}. \quad (54)$$

This term is block diagonal (it depends on the lattice site but not on the spin component). It is enough to define

$$[A(x)]_{cb}^{\alpha\beta} := -\frac{i c_{\text{sw}}}{16} (-1)^\alpha \delta_{\alpha\beta} \delta_{cb} [Q_{01}(x) - Q_{10}(x)] \quad (55)$$

on the finest level (before it was 0). This way, we preserve the form of the Dirac operator which we introduced previously and all the equations needed for the coarse gauge links remain the same.

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